

Análisis dimensional para ΔP ; pérdida de presión en el tubo.
 Pi de Buckingham (presentado por Genzel y Cimbala).

$$\Delta P = f(\rho, \mu, V, D, L, D_c, \Delta z) \quad M, L, T = j = 3$$

$$\left. \begin{array}{l} \Delta P [\text{Pa}] : ML^{-1}T^{-2} \\ \rho [\text{kg/m}^3] : ML^{-3} \\ \mu [\text{Pa} \cdot \text{s}] : ML^{-1}T^{-1} \\ V [\text{m/s}] : LT^{-1} \\ D [\text{m}] : L \\ L [\text{m}] : L \\ D_c [\text{m}] : L \\ \Delta z [\text{m}] : L \end{array} \right\} \begin{array}{l} k = n - j = 8 - 3 = 5. \\ \text{Parámetros repetitivos: } \rho, V, D. \end{array}$$

$$\begin{aligned} \textcircled{1} \Delta P \pi_1 &= \Delta P \rho^a V^b D^c \dots \textcircled{a} \\ &= (ML^{-1}T^{-2})^a (ML^{-3})^a (LT^{-1})^b (L)^c \\ &= M^{1+a} L^{-2-3a} T^{-2-b} L^b \\ &= M^{1+a} T^{-2-b} L^{-1-3a+b+c} \rightarrow M^0 T^0 L^0 \end{aligned}$$

$$\left. \begin{array}{l} 1+a=0 \\ -2-b=0 \\ -1-3a+b+c=0 \end{array} \right\} a=-1, b=-2, c=0$$

~~en 2~~

en ③ $\Delta P \rho^{-1} V^{-2} D^0 = \pi_1$
 $\pi_1 = \frac{\Delta P}{\rho V^2}$

$$Eu = \frac{\Delta P}{\rho V^2}$$

② μ

$$\pi_2 = \mu \rho^a V^b D^c \dots \text{⑥}$$

$$\mu = ML^{-1}T^{-1}$$

$$\pi_2 (ML^{-1}T^{-1})(ML^{-3})^a (LT^{-1})^b (L)^c$$

$$= \cancel{M} \cancel{L}^{-1} \cancel{T}^{-1} \cancel{M}^a \cancel{L}^{-3a} \cancel{L}^b \cancel{T}^{-b} \cancel{L}^c$$

$$= M^{1+a} L^{-1-3a+b+c} T^{-1-b}$$

$$\begin{cases} 1+a=0 \\ -1-b=0 \\ -1+3a+b+c=0 \end{cases}$$

$$\rightarrow \begin{cases} a=-1 \\ b=-1 \\ c=-1 \end{cases}$$

$$\begin{aligned} -1+3-1+c &= 0 \\ c &= -1 \end{aligned}$$

en ⑥, $\pi_2 = \frac{\mu}{\rho V D} = \frac{1}{Re}$

$$\text{③ } \pi_3 = L \rho^a V^b D^c \quad L=[L]$$

$$\pi_3 \cdot \frac{L}{D}$$

Igualmente, $\pi_4 = \frac{D_c}{D} \circ \frac{D}{D_c}$, $\pi_5 = \frac{\Delta z}{D}$

$$\therefore \frac{\Delta P}{\rho V^2} = F\left(Re, \frac{L}{D}, \frac{D}{D_c}, \frac{\Delta z}{D}\right)$$

Dado que es un sistema con tubo en espiral, combinando π_2 y π_4 , $De = Re \sqrt{\frac{D}{D_c}}$